

FUNCTIONS OF THYROID HORMONES

- TH have 2 major effects on body:
 - i. To ↑se BMR
 - ii. To stimulate growth in children.

1. ACTION ON BMR

- **Thyroxine** ↑ses **BMR** in body tissues **except** brain, retina, spleen, testes, and lungs.
- ↑ses **BMR** by ↑sing oxygen consumption of tissues.
- **Calorigenic Action:** It is a calorigenic action (↑ses BMR).

2. ACTION ON PROTEIN METABOLISM

- ↑ses synthesis of protein in cells:

1. By ↑sing translation of RNA:

↑ ses translation of RNA



Ribosomes activated



More proteins synthesize

2. By ↑sing transcription of DNA to RNA:

TH stimulates transcription of DNA to RNA



Accelerates synthesis of proteins

3. By ↑sing activity of mitochondria:

TH ↑ses activity of mitochondria in most cells



↑ se production of ATP



Energy for cellular activities

4. By ↑sing activity of cellular enzymes:

- TH ↑ses activity of intracellular enzymes such as α -glycerophosphate dehydrogenase and oxidative enzymes.
- These enzymes accelerate metabolism of proteins & carbohydrates.

3. ACTION ON CARBOHYDRATE METABOLISM

1. Thyroxine ↑ses absorption of glucose from GI tract.
2. Enhance glucose uptake by cells (*accelerating transport of glucose through the cell membrane*).
3. ↑ses breakdown of glycogen into glucose.
4. Accelerates gluconeogenesis.

4. ACTION ON FAT METABOLISM

- Thyroxine ↓ses fat storage by mobilizing it from adipose tissues and fat deposits.
- Thyroxine ↑ses free fatty acid level in blood.

5. ACTION ON PLASMA AND LIVER FATS

- Thyroxine ↓ses cholesterol, phospholipids and triglyceride levels in plasma.
- Atherosclerosis: Hyposecretion of thyroxine.

- **Fatty liver:** Thyroxine ↑se deposition of fats in liver.

6. ACTION ON VITAMIN METABOLISM

- **Thyroxine** ↑ses formation of many enzymes.
- **Vitamins may be utilized** during formation of enzymes:
 - ↓
 - Vitamin deficiency possible during **hypersecretion of thyroxine**.

7. ACTION ON BODY TEMPERATURE

- **TH** ↑ses heat production in body (*by accelerating various cellular metabolic processes & ↑sing BMR*):
 - ↓
 - Called **Thyroid Hormone-Induced Thermogenesis**.
- **Hypersecretion of thyroxine:**
 - Body temperature ↑ses greatly
 - ↓
 - Resulting in **excess sweating**.

8. ACTION ON GROWTH

- **↑se in thyroxine secretion** accelerates the growth of body, especially in growing children.
- **Lack of thyroxine** arrests growth (*same time causes early closure of epiphysis*).
- **Thyroxine promotes growth and development of brain** during fetal life and first few years of postnatal life:
 - ↓
 - Leads to **mental retardation** (*if deficient*).

9. ACTION ON BODY WEIGHT

- ↑se in thyroxine secretion ↓ses the body weight & fat storage.
- ↓se in thyroxine secretion ↑se the body weight becoz of fat deposition.

10. ACTION ON BLOOD

- Thyroxine accelerates erythropoietic activity & ↑se blood volume.
- Hyperthyroidism: Polycythemia is common.

11. ACTION ON CARDIOVASCULAR SYSTEM

- Thyroxine ↑ses overall activity of CVS.

1. On Heart Rate

- Thyroxine ↑ses HR.
- Important clinical investigation for diagnosis of **hypothyroidism & hyperthyroidism**.

2. On Force of Contraction

- Thyroxine ↑ses force of contraction.
- In **hyperthyroidism / thyrotoxicosis**, heart may become weak due to excess activity & protein catabolism:
 - ↓
 - Patient dies of **cardiac decompensation**
 - ↓
 - Failure of heart to maintain adequate circulation associated with **dyspnea, venous engorgement & edema**.

3. On Blood Vessels

- Thyroxine causes vasodilation by ↑sing metabolic activities:
 - ↓
 - Large quantity of metabolites produced

- ↓
- Cause **vasodilation**.

4. On Arterial Blood Pressure

- ↑**se in rate and force of contraction of heart** & ↑se in blood volume and blood flow by influence of thyroxine:
 - Cardiac output ↑ ses
 - ↓
 - Systolic pressure ↑ ses & Diastolic pressure ↓ ses
 - ↓
 - **So, only the pulse pressure ↑ses.**

12. ACTION ON RESPIRATION

- **Thyroxine ↑ses rate & force of respiration.**
- **↑sed metabolic rate:**
 - ↓
 - ↑ses demand for oxygen & formation of excess CO₂
 - ↓
 - Stimulate respiratory centers to ↑se rate & force of respiration.

13. FUNCTION ON CNS

Essential for development and maintenance of CNS.

1. On Development of CNS

- **Thyroxine promotes development of brain** during fetal life.
- **Thyroid deficiency results in:**
 - Mental retardation
 - Defective myelination & abnormal development of synapses.

2. On Normal Function of CNS

- Thyroxine also ↑ses blood flow to brain.
- Hypersecretion of thyroxine (*excess stimulation of CNS*):
 - ↓
 - Person has **extreme nervousness**
 - ↓
 - Develop **psychoneurotic problems** (*anxiety complexes, excess worries or paranoid thoughts*).
- Hyposecretion of thyroxine leads to **lethargy and somnolence**.

14. ACTION ON SEXUAL FUNCTION

- In men:
 - Hypothyroidism leads to **complete loss of libido** (*sexual drive*).
 - Hyperthyroidism leads to **impotence**.
- In women:
 - Hypothyroidism causes **menorrhagia & polymenorrhea**.
 - Hyperthyroidism leads to **oligomenorrhea & sometimes amenorrhea**.